MONASH University

Practice Test 1 - FIT1045 2025

For the test you will need to bring writing material (pen/pencil), a photo id, and your student card or M-PASS.

The following details will help ensure that we can match your attempt to your account in OnTrack and get the test back to you in your applied class.

Student name: _____

Student ID: _____

Applied day and time: _____

For this test you:

- must work alone.
- must not use any reference materials.
- will have 45 minutes to complete it.
- must write all answers within this test booklet.
- must hand back your test booklet at the end of the test session.
- must not take any notes or remove any test materials from the test location.
- must place all smart devices within your bag and not use these during the test.
- must not leave in the last 10 minutes of the test.

Your test will be graded as Complete where everything is at the expected standard, Fix and resubmit if you have made some small mistakes you can correct in class, or Redo if you appear to be falling behind. For Test 1, if you get Fix or Redo you will need to correct your mistakes in your applied class. Results will be updated in OnTrack.

Official Use		
☐ Complete	☐ Fix and resubmit	☐ Redo

1) Write a sentence to define each of the programming terminology / statements listed below.

1: Variable

2: Constant

3: Expression

4: Method Call

5: Assignment Statement

6: If Statement

7: While Loop

8: Do While Loop

9: Break and Continue

10: Terminal

11: Shell

2) Highlight an example of each of the following in this code. For statements that cross multiple lines, make sure to highlight all of the code associated with the statement.
   - Comment
   - Method call
   - Expression
   - If statement
   - While loop
   - Variable declaration
   - Constant declaration

```
const int PLAYER_SPEED = 3;

// Location of the player
int playerX = 800 / 2;
int playerY = 600 / 2;

OpenWindow("Player Move", 800, 600);

while (!QuitRequested())
{
    if (KeyDown(KeyCode.RightKey) && playerX < 800)
    {
        playerX += PLAYER_SPEED;
    }

    ClearScreen(ColorWhite());
    FillCircle(ColorBlack(), playerX, playerY, 25);
    RefreshScreen(60);

    ProcessEvents();
}
WriteLine("Thanks for playing");
```

3) Write the C# code, in the same style as P2-4 (i.e., no method/class declarations), to output a random value (between 1 and 100) and then ask the user if they want a different value (validate to ensure they enter Y or N). This should repeat presenting different random numbers until the user chooses N. Then output the message "Nice try" a number of times equal to the random number chosen.
You may need these methods:
   - public static int Rnd(int max); // get a random number from 0 to max - 1
   - public static void WriteLine(...);
   - public static void Write(...);
   - public static string ReadLine();

```
using static SplashKitSDK.SplashKit;
using static System.Convert;
```

4) Hand execute the following code within the space below. Ensure you follow the process and format described for doing this in the Field Guide.

```
int a = 2;
int b = 6;
int c = 0;
bool flag = false;
int i;

a = a + 3;
b = b - 1;

if (a == b)
{
   flag = true;
   c = a * 2;
}
else
{
   flag = false;
   c = b - a;
}

if (flag)
{
   for (i = 0; i < a; i++)
   {
       c = c - 1;
   }
}
else
{
   for (i = 0; i < b; i++)
   {
       c = c + 2;
   }
}
```